

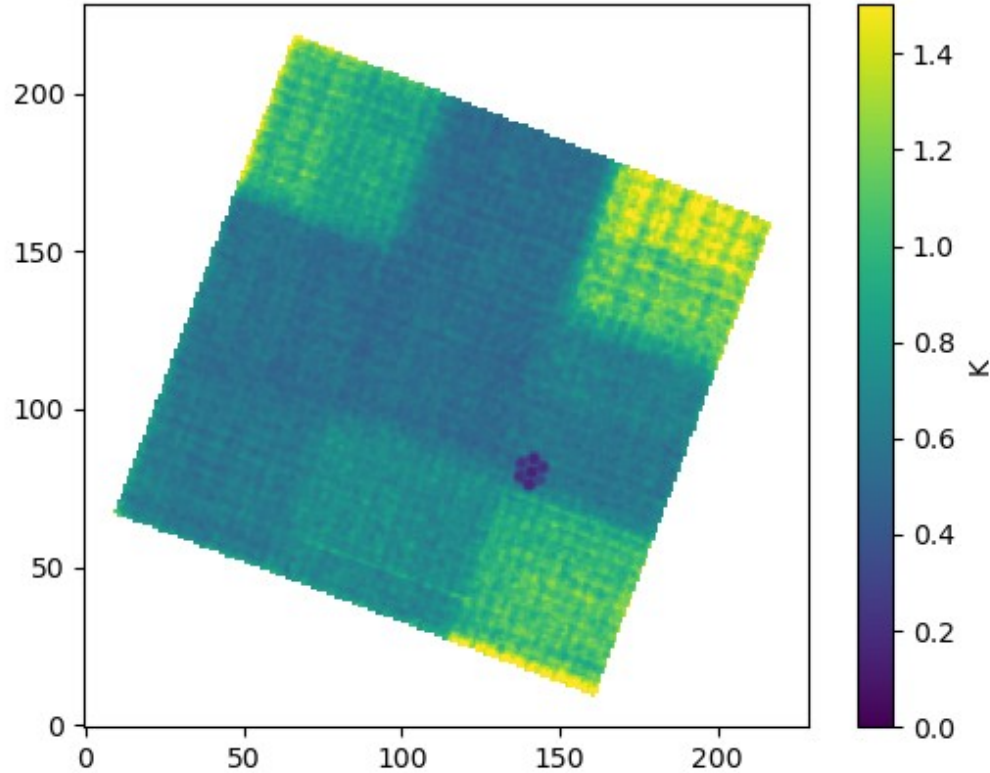
23 March 2026

https://github.com/Lim1029/W40-CII-Analysis/blob/main/gmm_cii.py

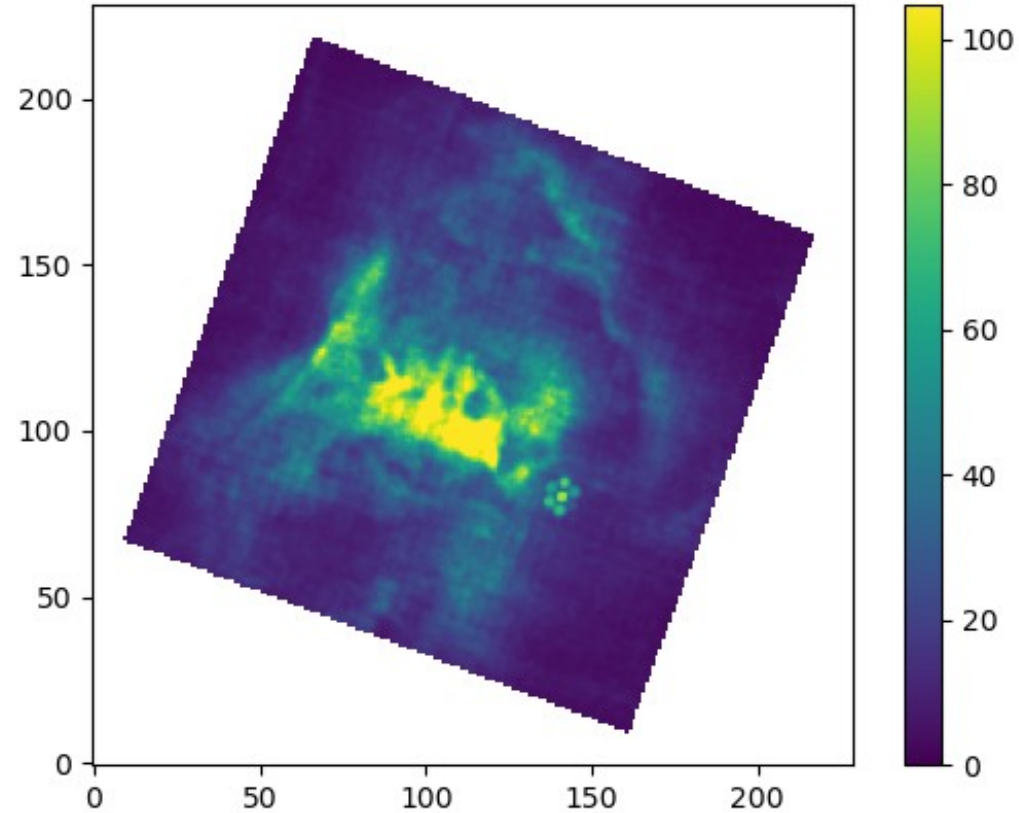
SOFIA W40 CII data

Using W40_CII_PCA_20_8_0p3_clean.fits

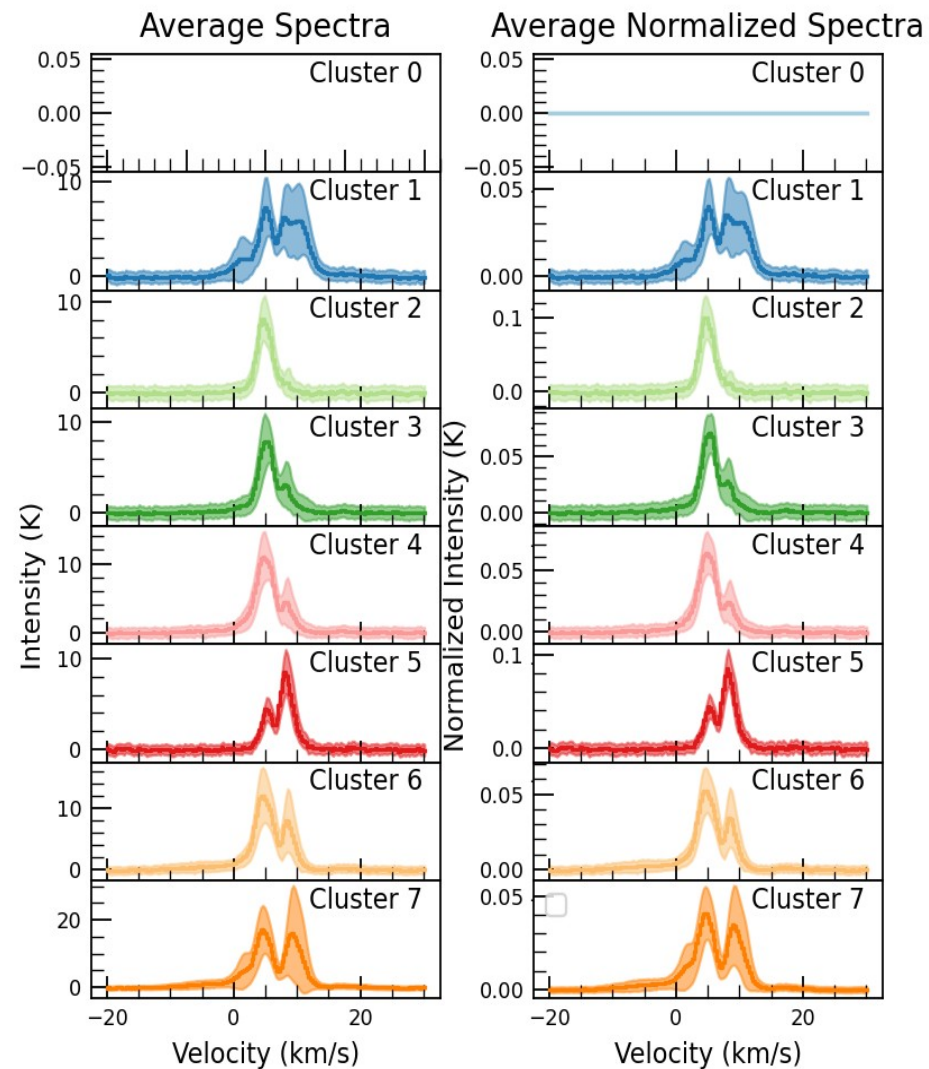
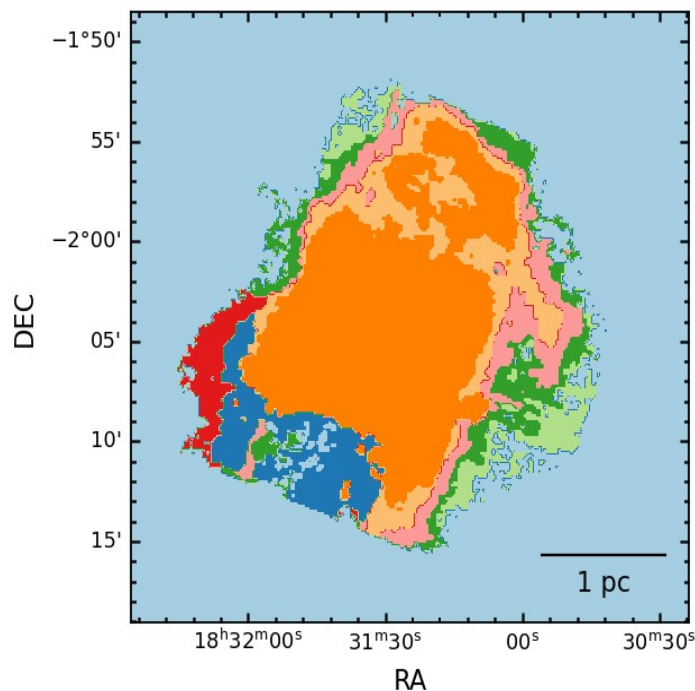
Noise distribution



SNR Map

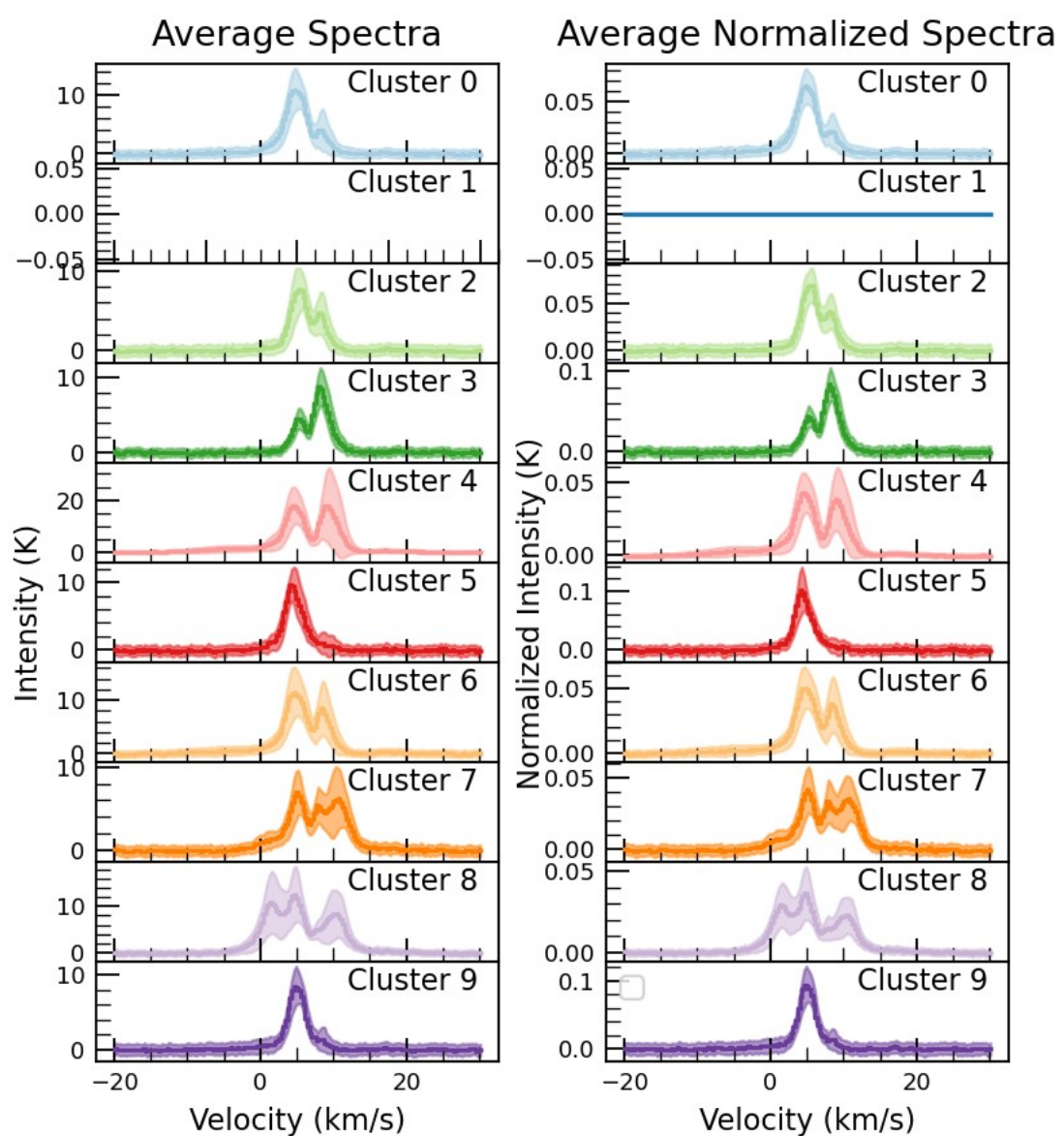
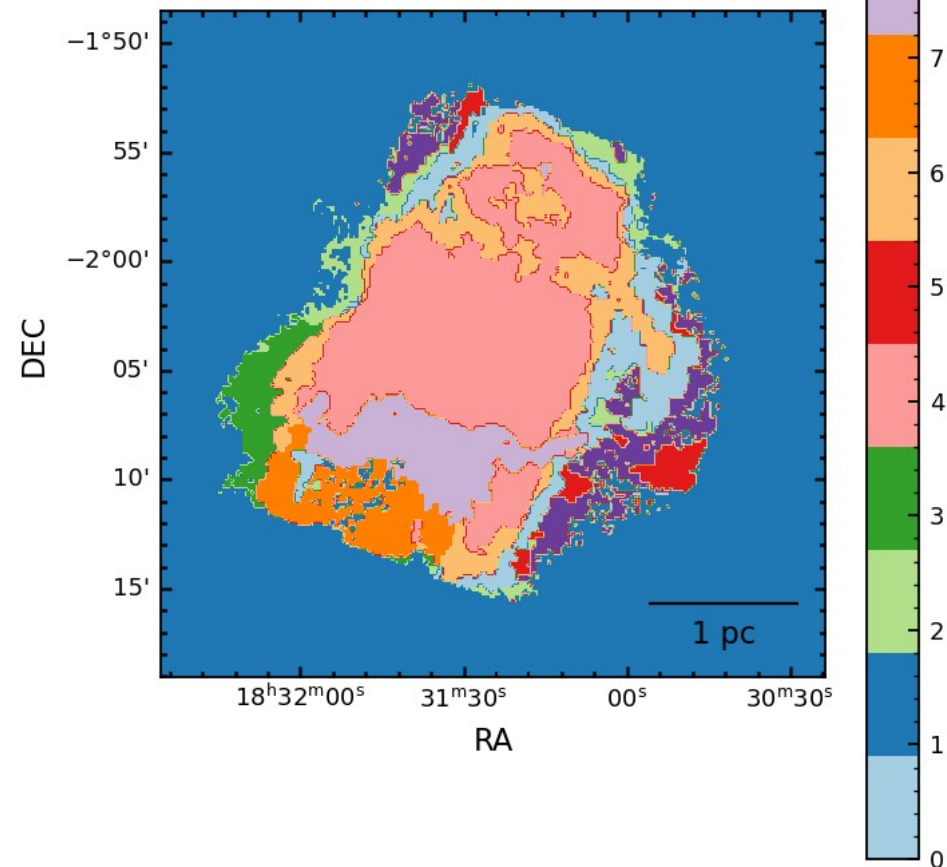


- Scikit-learn GMM
- Fixed 8 clusters
- Run on minmax normalized spectra
- Mask out spectra with $\text{SNR} < 10$



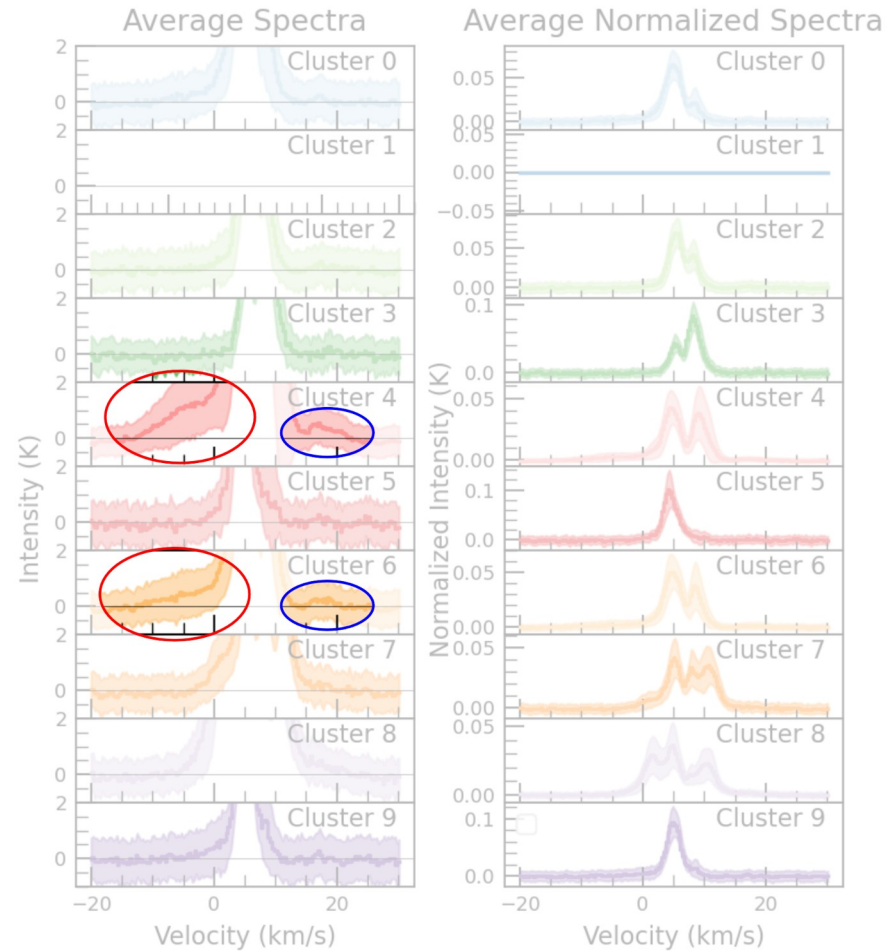
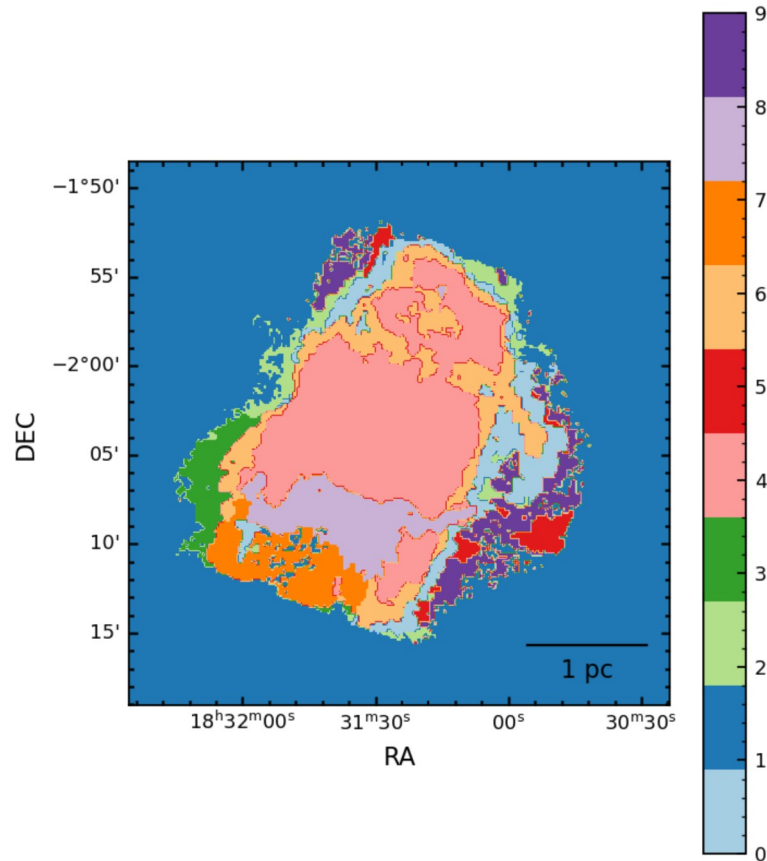
- Cluster 1 & 7 show rather high variations

- $N_{\text{comp}} = 10$
- Choosing different n_{comp} gives different clustering.
- How to decide the best n_{comp} value?



Blue expansion signature observed in 2 clusters (red circles)

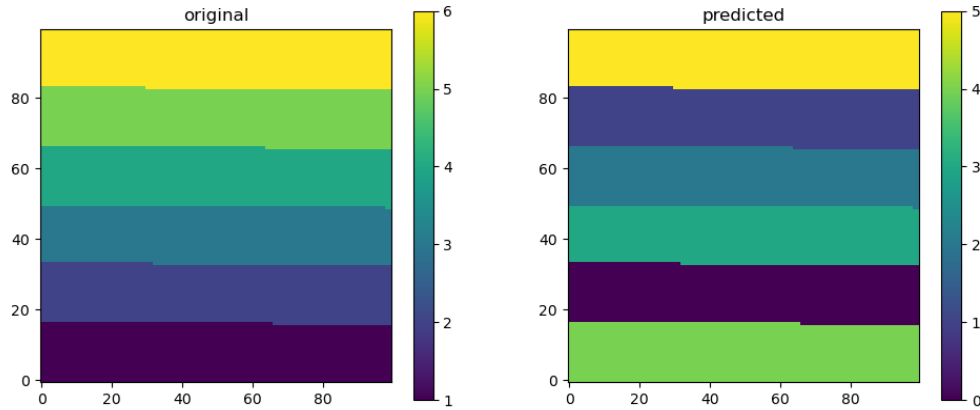
(Analysing on gmm with ncomp=10) Red expansion is weekly observed (green circles).



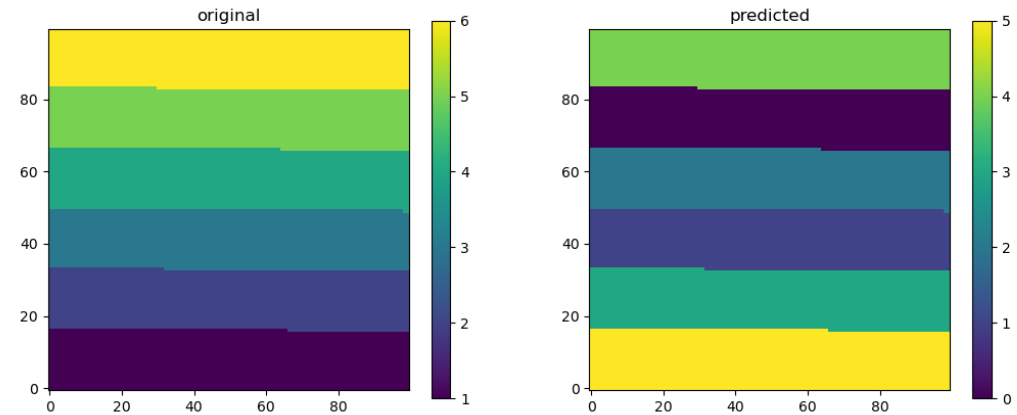
Scikit-learn GMM test

Reliability test of the algorithm on synthetic data

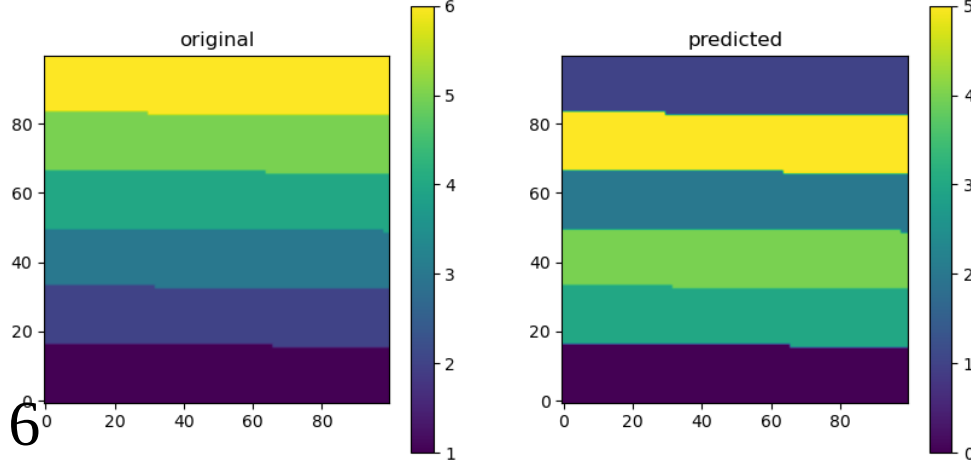
Synthetic data without noise



With uniform noise of constant rms



With noise with various rms



The algorithm seems to work with the presence of noise, but more robust test is needed

Selecting the best number of clusters (ncomp)

Still thinking how to do it....